LAYFIELD GEOSYNTHETICS

TECHNICAL SPECIFICATIONS

ENVIRO LINER® 4000 - GEOMEMBRANE

With its exceptional flexibility and endurance properties, Layfield's Enviro Liner® (EL) brand of geomembranes has been a trusted performer in our portfolio for many years. Available in four unique formulations, this series of geomembranes offers solutions for a wide range of containment applications.

Enviro Liner® 4000 is a specialized LLDPE polyethylene lining material that combines flexibility, UV stability, and chemical resistance, allowing for easy accommodation to differential settlement.

April 2023

Enviro Liner® 4000

Material Properties

Rev	ASTM	**Enviro Liner® 4020**	**Enviro Liner® 4030**	**Enviro Liner® 4040**	**Enviro Liner® 4060**
Thickness (minimum average)	D5199	20 mil 0.5 mm	30 mil 0.75 mm	40 mil 1.0 mm	60 mil 1.5 mm
Thickness (Lowest individual of 10 value)	D5199	-10%	-10%	-10%	-10%
Density (typical)	D792	≤ 0.939 g/ml	≤ 0.939 g/ml	≤ 0.939 g/ml	≤ 0.939 g/ml
Tensile Strength at Break	D6693	76 ppi 13 N/mm	114 ppi 20 N/mm	152 ppi 27 N/mm	228 ppi 40 N/mm
Elongation	D6693	800%	800%	800%	800%
Tear Resistance	D1004	11 lbs 50 N	16 lbs 70 N	22 lbs 100 N	33 lbs 150 N
Puncture Resistance	D4833	29 lbs 120 N	42 lbs 190 N	56 lbs 250 N	84 lbs 370 N
Carbon Black Content	D6370	≥ 2.0%	≥ 2.0%	≥ 2.0%	≥ 2.0%
Carbon Black Dispersion	D5596	Cat 1 or 2	Cat 1 or 2	Cat 1 or 2	Cat 1 or 2
UV Resistance HPOIT retained after 1600 hrs	D7238 D5885	35%	35%	35%	35%

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For up-to-date technical information, be sure to visit us online at www.LayfieldGroup.com

ENVIRO LINER® 4000 - GEOMEMBRANE

April 2023

Enviro Liner® 4000 Minimum Shop Seam Strengths

Style	ASTM D6392	**Enviro Liner® 4020**	**Enviro Liner® 4030**	**Enviro Liner® 4040**	**Enviro Liner® 4060**
Heat Bonded Seam Strength	25.4 mm (1”) Strip	30 ppi 5.2 N/mm	45 ppi 7.7 N/mm	60 ppi 10.3 N/mm	Not shop fabricated
Peel Adhesion Strength (Wedge Welded	25.4 mm (1”) Strip	25 ppi 4.3 N/mm	38 ppi 6.5 N/mm	50 ppi 8.6 N/mm	Not shop fabricated

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Enviro Liner® 4000 Minimum Field Seam Strengths

Style	ASTM D6392	**Enviro Liner® 4020**	**Enviro Liner® 4030**	**Enviro Liner® 4040**	**Enviro Liner® 4060**
Heat Bonded Seam Strength	25.4 mm (1”) Strip	30 ppi 5.2 N/mm	45 ppi 7.7 N/mm	60 ppi 10.3 N/mm	90 ppi 15.7 N/mm
Peel Adhesion Strength (Wedge Welded	25.4 mm (1”) Strip	22 ppi 3.8 N/mm	34 ppi 5.9 N/mm	44 ppi 8.7 N/mm	66 ppi 11.6 N/mm

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